Predictive Forecasting of Care Load and Placement Demand

A Time-Series and Machine-Learning Study of the HHS Unaccompanied Alien Children (UAC) Program

Abhinav Ray
Unified Mentor Project

October 15, 2026

Abstract

The U.S. Department of Health and Human Services' Unaccompanied Alien Children (UAC) Program publishes a high-quality daily time series describing the flow of children through federal custody, yet operational decisions have historically relied on descriptive reporting rather than forward-looking forecasts. This paper develops and evaluates a forecasting pipeline for two operationally critical quantities: the number of children in active HHS care, and the daily discharge (placement) rate. We audit and clean 720 raw daily records (2023-01-12 to 2025-12-21), reindex to a continuous 1,075-day daily series, and identify a sharp structural regime break around February 2025 in which every series collapses to a smaller, far more stable operating range. We compare naive baselines, classical statistical models (ARIMA, Holt-Winters exponential smoothing), and machine-learning regressors (Random Forest, Gradient Boosting) using strict time-based walk-forward validation across 1-, 7-, and 14-day horizons. Damped-trend Holt-Winters exponential smoothing is the strongest model at every horizon beyond one day, while tree-based ML models underperform substantially, a result we attribute to the regime break limiting their ability to specialize on the current operating range. We define and compute four operational KPIs (forecast accuracy, capacity breach probability, surge lead time, forecast stability), quantify forecast uncertainty via simulation-based prediction intervals, and deliver the results through a public interactive dashboard. We close with concrete recommendations and limitations for HHS operational planners.

Contents

1 Introduction **2**
1.1 Background and Context . . . 2
1.2 Problem Statement . . . 2
1.3 Objectives . . . 2

2 Data Description **3**

3 Methodology **3**
3.1 Data Audit and Cleaning . . . 3
3.2 Time-Series Preparation . . . 3
3.3 Exploratory Data Analysis . . . 4
3.3.1 Regime Break . . . 4
3.3.2 Decomposition and Stationarity . . . 4

3.3.3 Flow Balance 4
3.4 Feature Engineering 4
3.5 Train–Test Strategy 4
3.6 Forecasting Models 5

4 Results **5**
4.1 Model Comparison 5
4.2 Why Machine Learning Underperformed 6
4.3 Discharge Demand Forecast 6
4.4 Uncertainty Quantification 6
4.5 Key Performance Indicators 6

5 Streamlit Web Application **7**

6 Limitations **7**

7 Recommendations **8**

8 Conclusion **8**

1 Introduction

1.1 Background and Context

The UAC Program operates in a high-uncertainty environment in which sudden changes in border activity, policy enforcement, or humanitarian conditions can rapidly increase the number of children entering federal care. While descriptive analytics can explain what has already happened, HHS decision-makers require forward-looking intelligence to answer three operational questions:

- How many children will be under HHS care in the coming days or weeks?
- Will discharge (placement) capacity be sufficient to offset incoming transfers?
- When should shelters, medical staff, and caseworkers be scaled up in advance?

This project introduces predictive modeling to enable proactive, rather than reactive, child-welfare and healthcare resource planning.

1.2 Problem Statement

Despite having high-quality daily time-series data, the UAC Program currently lacks short-term forecasts of children in HHS care, predictive estimates of discharge demand, and early-warning indicators of upcoming capacity stress. As a result, operational responses are often delayed, increasing overcrowding risk, staff burnout, and length of stay for children in care.

1.3 Objectives

Primary objectives:

1. Forecast the number of children in HHS care.
2. Estimate the future imbalance between intake and exits.

3. Predict short-term discharge demand.

Secondary objectives:

1. Provide early warnings for healthcare planners.
2. Quantify forecast uncertainty.
3. Compare statistical and machine-learning forecasting approaches.

2 Data Description

The dataset consists of daily counts across five columns, spanning 2023-01-12 to 2025-12-21 (720 raw daily reports):

Column	Description
Date	Reporting date
Children apprehended and placed in CBP custody	Daily intake volume
Children in CBP custody	Active CBP care load
Children transferred out of CBP custody	Flow into the HHS system
Children in HHS Care	Active HHS care load (primary forecasting target)
Children discharged from HHS Care	Successful sponsor placements (secondary target)

3 Methodology

3.1 Data Audit and Cleaning

Initial inspection revealed two data-quality issues that required resolution before any time-series work could proceed:

- **Apparent duplicate dates** (449 of 720 rows): investigation showed these were in fact blank trailing rows with `NaT` dates introduced by CSV export artifacts, not genuine duplicate reports. All `NaT` rows were dropped, leaving 720 valid daily records with zero true duplicates.
- **Incorrect dtype**: the *Children in HHS Care* column was read as text due to comma thousand-separators (e.g. `"7,546"`). This was corrected by stripping separators and casting to numeric before any further analysis.

3.2 Time-Series Preparation

The cleaned 720-row series was reindexed onto a continuous daily calendar spanning the full date range, exposing 355 missing calendar days (approximately 85% of which fall on Fridays and Saturdays, consistent with a weekend/holiday reporting gap rather than missing operational activity). Gaps were filled via linear interpolation, producing a continuous 1,075-day daily series; the longest single gap was 9 days.

3.3 Exploratory Data Analysis

3.3.1 Regime Break

The single most consequential finding of the exploratory phase is a sharp structural break around **February 2025**, visible simultaneously across all five series (Figure 1). Before the break, *Children in HHS Care* averaged approximately 7,700 with a peak of 11,516 (2023-12-20); after the break it averaged approximately 2,300 and has remained within a narrow 1,972–2,528 band since March 2025 (trough 1,972 on 2025-08-21). Median absolute daily change fell from approximately 57 to approximately 6, indicating the post-break regime is both lower in magnitude and dramatically less volatile. All downstream modeling and evaluation treats this break as a first-class consideration rather than noise.

3.3.2 Decomposition and Stationarity

STL decomposition (period = 7, robust fitting) of *Children in HHS Care* shows the trend component carrying the majority of the signal; the weekly seasonal component is approximately ±200 in the pre-break period and fades to near zero after the break (Figure 2). An Augmented Dickey-Fuller test confirms the series is non-stationary at the level ($p = 0.66$) but stationary after first differencing ($p \approx 1.18 \times 10^{-5}$), establishing $d = 1$ for subsequent ARIMA specification (Figure 3).

3.3.3 Flow Balance

Net pressure, defined as daily transfers into HHS custody minus daily discharges, averages approximately −55 children per day across the full series and is positive (i.e. the system is filling faster than it is emptying) on only about 31% of days. Net pressure correlates only weakly ($r \approx 0.35$) with the day-over-day change in HHS care load, indicating it is a useful but not dominant standalone predictive signal (Figure 4). Discharges also show a weekly pattern, running higher on Thursday, Friday, and Sunday than on Monday through Wednesday in the pre-break period.

3.4 Feature Engineering

For the machine-learning models, the following features were derived, with every feature constructed to use only information available strictly before the day being predicted (lag-1 shifting applied to all rolling statistics to prevent leakage):

- **Lag features**: values at $t-1$, $t-7$, and $t-14$.
- **Rolling statistics**: 7-day and 14-day rolling mean and standard deviation.
- **Flow-based signal**: net pressure (transfers minus discharges), lagged by one day.
- **Calendar effects**: day of week, month.

3.5 Train–Test Strategy

All models share a common evaluation protocol:

- A strict chronological split, with the final 90 days of the series (2025-09-22 to 2025-12-21, entirely within the post-break regime) held out as the test window.
- **Walk-forward validation** for the statistical models: the model is refit on an expanding history window and forecasts are generated at a 7-day step cadence through the test period.

- **Multi-horizon evaluation** at 1-, 7-, and 14-day horizons for every model.
- Machine-learning models are trained as *direct multi-horizon* regressors (one model fit per horizon) rather than recursively, to avoid compounding single-step errors over longer horizons.

3.6 Forecasting Models

- **Baselines**: naive persistence (tomorrow equals today), and a 7-day moving average.
- **Statistical models**: ARIMA(2,1,2), and Holt-Winters exponential smoothing with an additive, damped trend.
- **Machine-learning models**: Random Forest Regressor and Gradient Boosting Regressor, each trained separately per forecast horizon.

4 Results

4.1 Model Comparison

Table 1 reports Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) for every model at every horizon, evaluated on the held-out post-break test window.

Table 1: Model comparison across forecast horizons (sorted by MAE within each horizon).

Horizon	Model	MAE	RMSE	MAPE (%)
1-day	ARIMA(2,1,2)	5.23	5.91	0.23
	Holt-Winters	5.24	6.43	0.23
	Naive persistence	5.75	6.40	0.25
	Random Forest	9.42	11.35	0.41
	Gradient Boosting	18.93	22.16	0.83
	Moving average	20.47	21.63	0.89
7-day	**Holt-Winters**	**10.31**	**13.41**	**0.45**
	Naive persistence	14.43	18.16	0.63
	ARIMA(2,1,2)	15.89	19.82	0.69
	Random Forest	28.27	32.56	1.22
	Moving average	29.88	32.36	1.30
	Gradient Boosting	92.26	98.01	3.97
14-day	**Holt-Winters**	**17.70**	**23.04**	**0.77**
	Naive persistence	28.18	34.33	1.22
	ARIMA(2,1,2)	29.78	35.86	1.29
	Moving average	43.11	48.16	1.87
	Gradient Boosting	135.39	140.31	5.79
	Random Forest	153.77	170.54	6.49

Key finding: damped-trend Holt-Winters exponential smoothing is the strongest model at every horizon beyond one day. At the 1-day horizon, ARIMA and Holt-Winters are statistically indistinguishable and both narrowly beat naive persistence. At 7 and 14 days,

Holt-Winters pulls decisively ahead; notably, **ARIMA underperforms the naive baseline at both the 7- and 14-day horizons**, indicating its fixed order is not well tuned for longer-range drift in this series.

4.2 Why Machine Learning Underperformed

Random Forest and Gradient Boosting underperform every statistical model at every horizon, and the gap widens sharply with horizon length – Gradient Boosting is roughly 9× worse than Holt-Winters at the 7-day horizon and both tree models are roughly 8× worse at 14 days (Figure 6). We attribute this to the regime break: both tree ensembles are trained on data spanning both the pre- and post-break regimes, and their leaf-level predictions are not well specialized to the narrow, low-volatility post-break range that defines the test window. Statistical models, by contrast, adapt their internal level and trend state directly from the most recent observations, allowing them to track the current regime without an explicit regime indicator feature. Feature importance analysis (Figure 5) confirms that the 1-day lag dominates predictive power for the ML models, consistent with how stable and autocorrelated the post-break series is.

4.3 Discharge Demand Forecast

A second Holt-Winters model was fit to forecast daily discharges from HHS care. An initial version of this model, trained on data spanning both regimes, produced an unrealistic extrapolated trend (forecast mean climbing from approximately 19 to nearly 69 children/day by day 14, well above the historical maximum of 39). Restricting training data to the post-break period alone resolved this: the corrected model's 14-day forecast mean converges to approximately 14, closely matching the recent historical mean of 14.9 (Figure 8). This episode illustrates a broader lesson from this project: a regime break does not only degrade machine-learning models (Section 4.2); it can equally distort a statistical model's trend estimate if the model is naively fit across the full history rather than the operationally relevant regime.

4.4 Uncertainty Quantification

Prediction intervals for the care-load forecast were generated via Monte Carlo simulation (500 repetitions) from the fitted Holt-Winters model, taking empirical 5th/95th percentiles at each horizon day. The interval widens substantially with horizon: at day 1 the 90% interval spans approximately 2,021–2,251 (width $\approx$ 230); by day 14 it spans approximately 572–3,697 (width $\approx 3,125$) – a roughly 13× widening (Figure 9). This widening is the central practical constraint on how far ahead the forecast should be trusted for operational decisions.

4.5 Key Performance Indicators

Four KPIs were defined and computed to translate forecasting performance into operationally meaningful terms:

The capacity breach probability rises with horizon, consistent with the widening uncertainty interval discussed above, providing planners with a horizon-dependent early-warning signal even though current care-load levels sit comfortably below the chosen capacity threshold. The surge lead time KPI could not be estimated from the 90-day test window because no surge-threshold crossing occurred during this unusually stable post-break period; this is reported as a limitation

Table 2: Key Performance Indicators (7-day horizon unless noted).

KPI	Definition	Value
Forecast Accuracy (%)	100 − MAPE	99.55%
Capacity Breach Probability	P(simulated forecast > capacity threshold), threshold set at 3,000 (above the post-break observed range)	4.8%
Surge Lead Time	Median number of days the forecast exceeds a rolling-mean-plus-2σ surge threshold before the actual surge occurs	Not estimable in this window
Forecast Stability Index	$1 - (\sigma_{\text{fold MAE}}/\mu_{\text{fold MAE}})$ across walk-forward folds	0.41

of the evaluation window rather than a model failure. The forecast stability index of 0.41 indicates moderate, not high, consistency of forecast error across walk-forward folds, and should be interpreted alongside the MAE/RMSE/MAPE table rather than in isolation.

5 Streamlit Web Application

The forecasting pipeline is delivered as a live, interactive dashboard (https://unified-uac-project.streamlit.app/; source code at https://github.com/abhinavray07/UAC_project) implementing all required core modules and user capabilities:

- **Core modules**: future care-load forecast chart, discharge demand forecast panel, model selection and comparison table, confidence interval visualization.
- **User capabilities**: forecast horizon selector (1/7/14 days), model toggle, and a baseline-versus-capacity-stress scenario comparison view.

All forecasts, intervals, and KPIs are precomputed in the analysis notebooks and loaded by the application at runtime; no model is retrained interactively, ensuring consistent, reproducible figures for stakeholders.

6 Limitations

- **High accuracy reflects an unusually stable period.** The ∼99% accuracy figures are a consequence of how stable the post-break regime has been (median daily change of single digits); they should not be read as a guarantee of performance should a new surge or regime shift occur, which is precisely the scenario this project is intended to help anticipate.
- **Surge lead time is unvalidated.** No genuine surge event occurred in the 90-day test window, so this KPI's real-world reliability remains untested; validating it would require either a longer evaluation window or back-testing against the volatile pre-break period.
- **Regime sensitivity of trend-based models.** As shown by the discharge-model trend-extrapolation issue (Section 4.3), any model with a trend component is sensitive to how much pre-break data it is trained on. All final models in this study were refit on post-break data

only, but any future retraining must preserve this practice, or re-evaluate it, if further regime shifts occur.

- **Machine-learning models are not currently production-ready** for this series given their large errors at longer horizons; they are retained in the comparison for completeness and as a baseline for future work (e.g., explicit regime-indicator features, or regime-segmented training).

7 Recommendations

1. **Adopt Holt-Winters (damped trend) as the production forecasting model** for both care load and discharge demand, trained on post-break data, with scheduled retraining as new data accrues.
2. **Treat forecasts beyond approximately 7 days as directional rather than precise**, given the roughly 13× widening of the prediction interval between day 1 and day 14.
3. **Monitor the capacity breach probability at the 7- and 14-day horizons** as a standing early-warning indicator; a sustained rise in this figure should trigger a staffing and shelter-capacity review well before the breach itself occurs.
4. **Re-validate the surge lead time KPI** against a longer or more volatile historical window before relying on it operationally, since the current test period contained no genuine surge event.
5. **Re-run the full modeling pipeline promptly after any future regime shift** is suspected, rather than waiting for a scheduled retraining cycle, given how sensitive both the ML and statistical models were shown to be to training-data regime composition.

8 Conclusion

This project elevates the UAC dataset from historical reporting to predictive intelligence. A rigorous data audit uncovered and resolved two data-quality issues and one major structural regime break; a systematic comparison of six forecasting approaches across three horizons identified damped-trend Holt-Winters exponential smoothing as the most reliable model for both care-load and discharge-demand forecasting; and the resulting forecasts, uncertainty intervals, and KPIs were delivered through a live, interactive dashboard built for non-technical stakeholders. By applying this pipeline operationally, HHS planners can move from reactive to proactive resource allocation, anticipating capacity stress days in advance rather than discovering it after the fact.

References

- Seabold, S., & Perktold, J. (2010). *statsmodels: Econometric and statistical modeling with Python.* Proceedings of the 9th Python in Science Conference.
- Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python.* Journal of Machine Learning Research, 12, 2825–2830.
- Streamlit Inc. *Streamlit Documentation.* https://docs.streamlit.io

- U.S. Department of Health and Human Services, Office of Refugee Resettlement. *Unaccompanied Alien Children Program Data*.

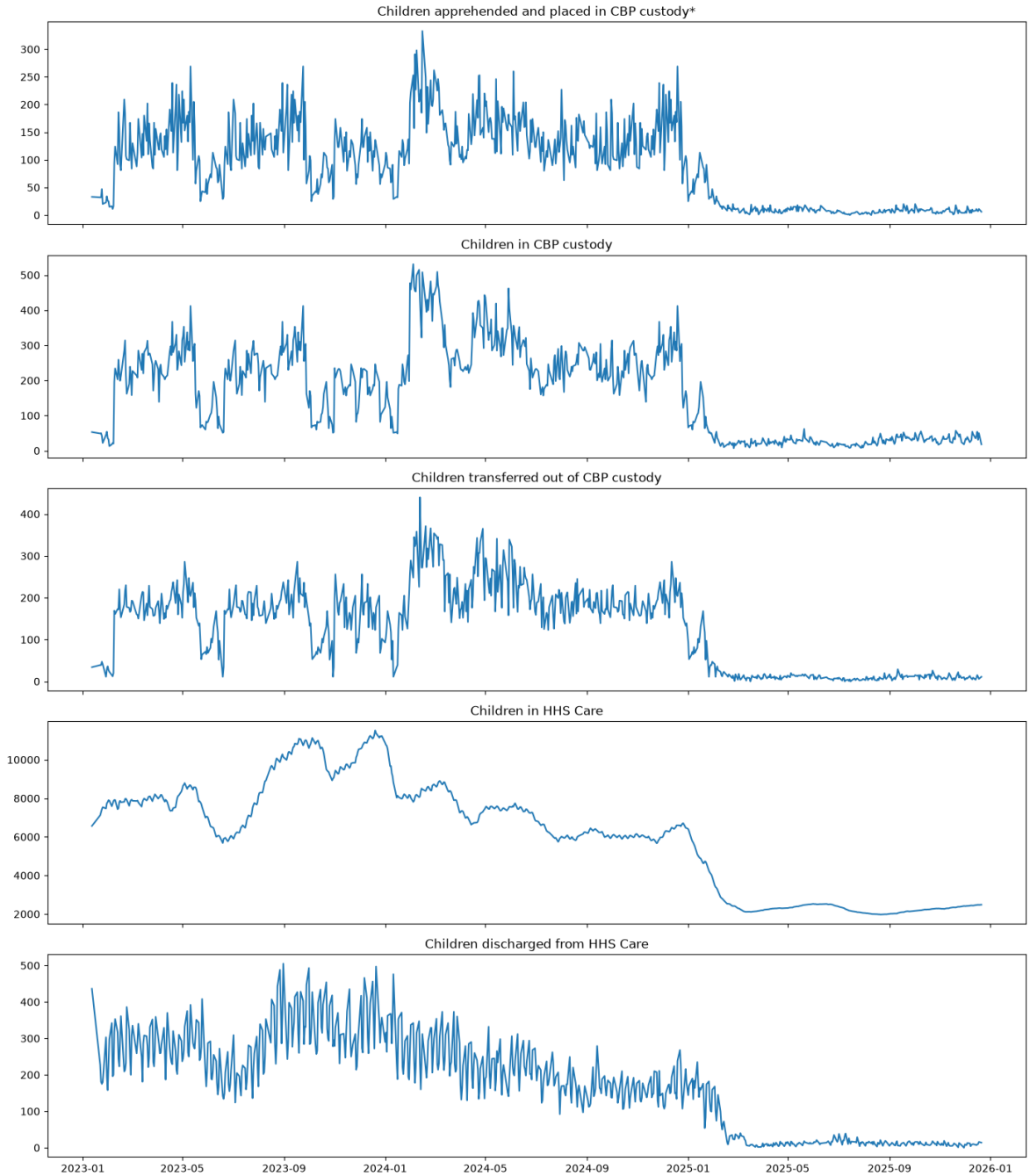


Figure 1: All five daily series after gap-filling, showing the synchronized regime break across intake, custody, and discharge flows.

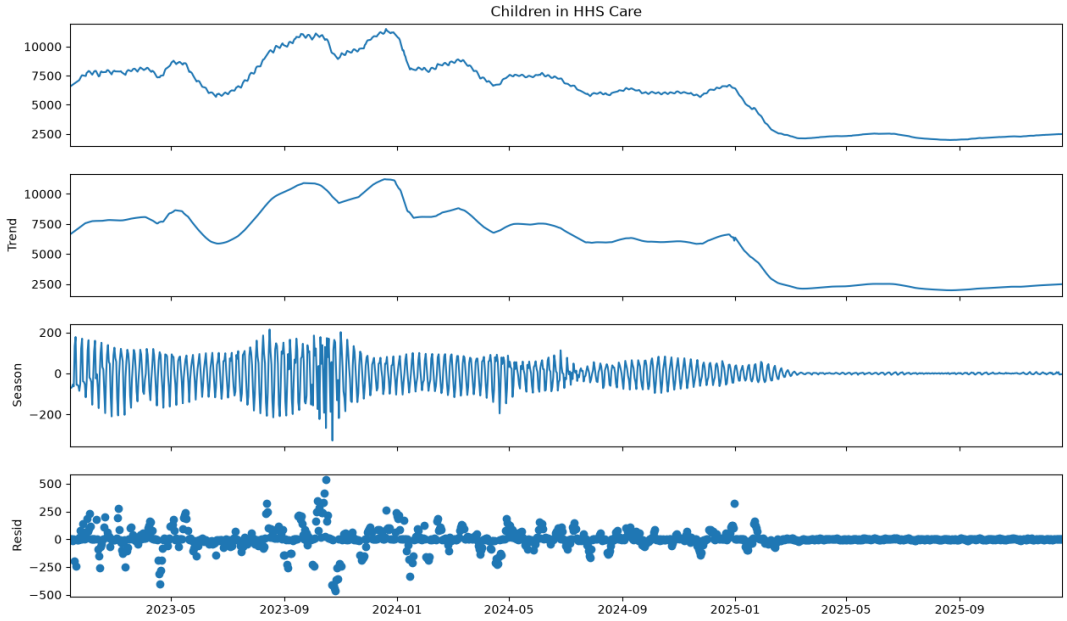


Figure 2: STL decomposition (period=7) of Children in HHS Care into trend, seasonal, and residual components.

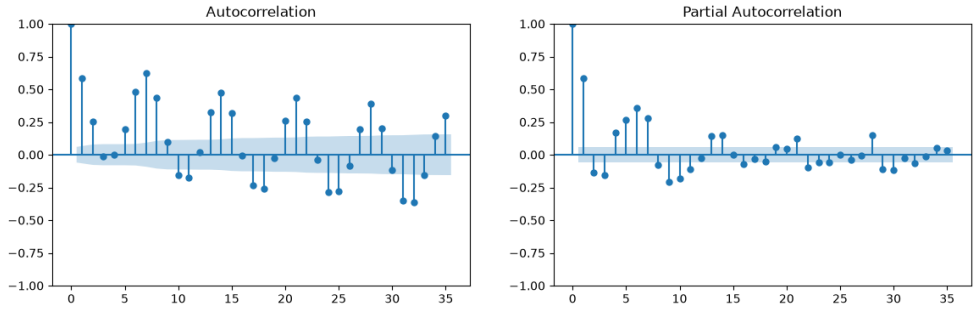


Figure 3: ACF and PACF of the first-differenced series, used to inform ARIMA order selection.

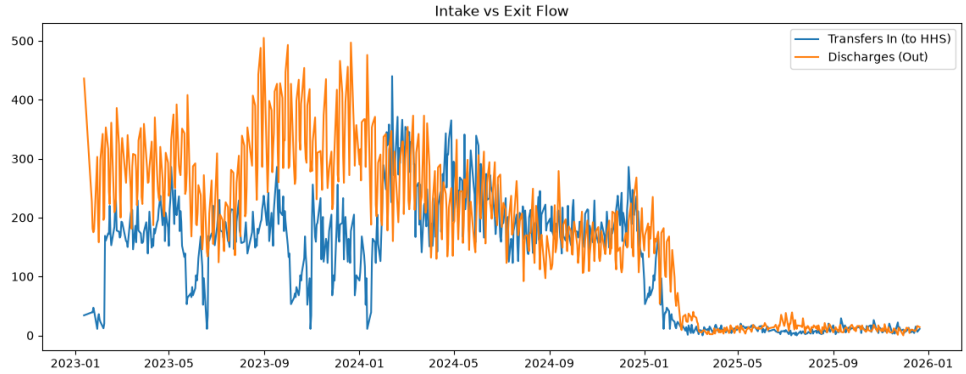


Figure 4: Daily transfers into HHS custody versus daily discharges from HHS care.

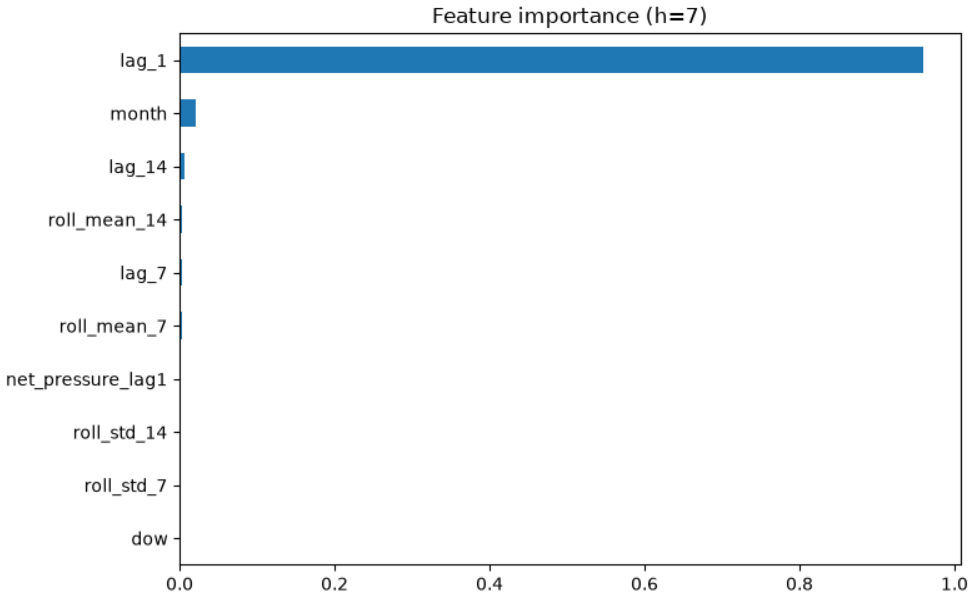


Figure 5: Gradient Boosting feature importance at the 7-day horizon.

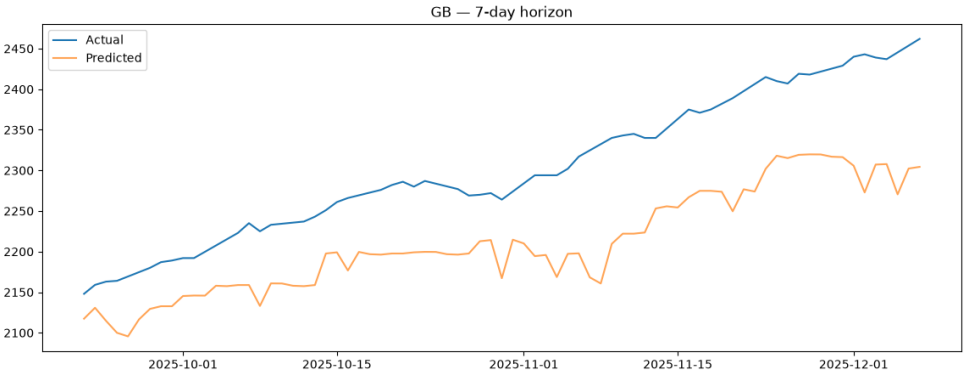


Figure 6: Gradient Boosting forecast versus actual, 7-day horizon, illustrating degradation on the post-break test window.

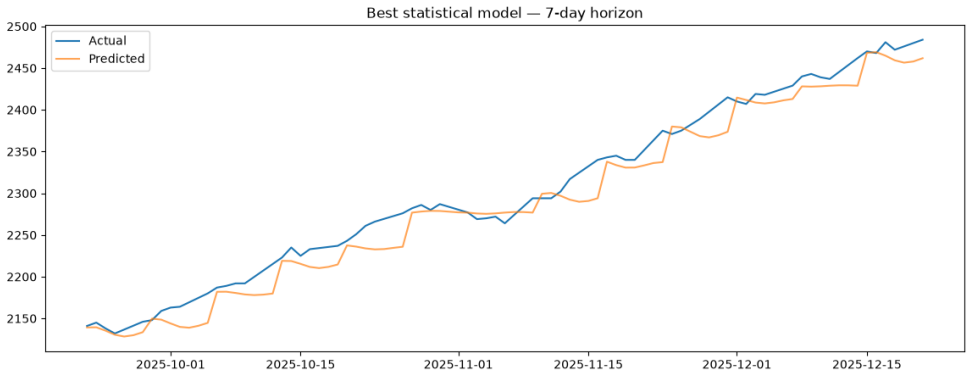


Figure 7: Best statistical model forecast versus actual, 7-day horizon.

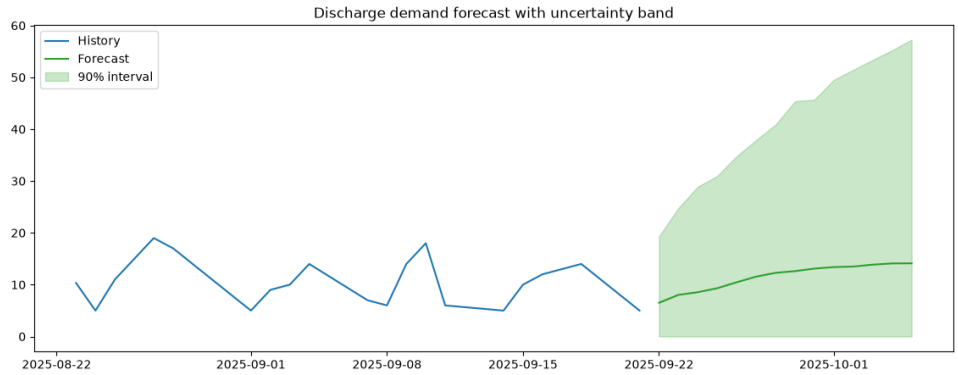


Figure 8: 14-day discharge demand forecast with 90% prediction interval, post-break-trained model.

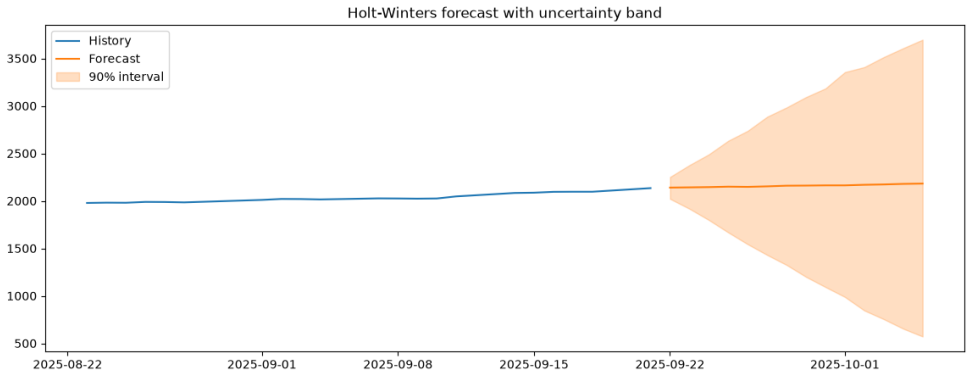


Figure 9: 14-day care-load forecast with 90% prediction interval (Holt-Winters, simulation-based).